

OceanCube 3-D

The reconstruction as a three-dimensional body of water - where the gaps are the sea floor, not missing data.

PORT	PAGE FILE	MODULE	FALLBACK
8504	app/phase2/cube_page.py	phase2/cube/ocean_cube.py	2-D slice if 3-D is unavailable

A profile is one line. A map is one depth. The cube is the whole reconstructed volume at once - and it is the fastest way to make a non-specialist audience understand that this project outputs a body of water, not a chart.

1. What this feature is, in plain words

The model produces a temperature at every one of 24,000 surface cells and at each of 15 depths. Stack those and you have a solid block of ocean. This page draws that block and lets you rotate it.

The single most important thing on the page is what the holes mean. Large parts of the basin are shallower than 1000 m, so those cells simply have no water at the deeper levels. Roughly a quarter of ocean cells never reach 1000 m at all. Those gaps are the sea floor - they are a fact about the Earth, not a failure of the model - and the page is built to say so rather than let a viewer assume the model gave up there.

The vertical axis is deliberately exaggerated, and the caption states by how much. One kilometre of depth stretched across sixty degrees of longitude is a film of water thinner than a sheet of paper; drawn honestly to scale you would see nothing at all.

THE EVERYDAY PICTURE

It is an MRI scan of a sea. A single slice is useful to a specialist; the rotating volume is what makes everyone else in the room understand what was actually built.

2. What you actually see on the screen

The good case and the safe case, in the same page:

- A rotatable 3-D volume of reconstructed temperature across the whole basin and all 15 depths.
- Gaps where the sea floor rises above the depth being drawn, labelled as bathymetry and not as missing output.
- A stated vertical exaggeration factor, so nobody reads the shape as a true aspect ratio.
- A coarser horizontal sampling for the renderer, with the caption saying the science is unchanged and only the picture is coarser.
- An automatic fallback to a flat 2-D depth slice, with the reason printed, if the 3-D renderer is missing or the volume is too large for a browser.

3. How to run it, step by step

RUN THIS

```
PYTHONPATH=src python -m streamlit run app/phase2/cube_page.py --server.port 8504
```

- 1 Run the command and open **http://localhost:8504**.
- 2 Pick a date inside the model's bundle window.
- 3 Let the volume build, then drag to rotate. Turn it until the audience is looking up from below - the shape of the continental shelf appears immediately.
- 4 Point at a hole and say the sentence out loud: that is the sea floor, not missing data.
- 5 Change the vertical exaggeration and let the audience watch the basin flatten. This is the moment they understand the true proportions of the problem.
- 6 If the 3-D view is unavailable on the demo machine, the page draws the 2-D slice and prints why - use that as an example of designed-in graceful degradation.

WHY THE FALLBACK EXISTS

The 3-D view needs an optional plotting library that is not guaranteed on a demo machine, and this project has already had a page fail that way once on this very laptop. A plainer chart that always renders beats a beautiful one that might not.

4. Why the problem statement needs this

The problem statement asks for a depth-resolved reconstruction across a region, not a point forecast. The cube is the direct visual proof that the output really is three-dimensional and complete - every ocean cell, every standard depth, one date.

It also demonstrates a discipline that runs through the whole project: an absence is never drawn as a value. A cell below the sea floor is refused rather than filled, because a filled cell would silently assert that water exists where there is rock.

PS requirement	How this feature answers it
Depth-wise reconstruction across the region	Renders all 15 depths across all 24,000 grid cells as a single object.
0.25 degree spatial resolution	The volume is built on the frozen 100 x 240 grid; only the on-screen sampling is coarsened.
Proof of concept over BoB and Arabian Sea	Both basins are visible in one view, with the shelf structure of each clearly different.

5. The measured numbers behind it

The classification behind the picture is arithmetic that must add up exactly, and it does [VERIFIED]:

Check	Numbers	Why it matters
Cell classification at 100 m	9,763 water + 2,069 below sea floor + 12,168 land = 24,000	Exactly the grid size. No cell is unclassified or double-counted.
Cell classification at 1000 m	8,973 water + 2,859 below sea floor + 12,168 land	Land is unchanged and water is strictly fewer, as physics requires.
Ocean cells at both depths	water + sea floor = 11,832 at both	The same ocean-cell count the export timing artifact independently records.
Cells with water at all 15 levels	8,973	About 24% of ocean cells never reach 1000 m - which is what the gaps in the cube are.

6. The impact - and who it is for

Who	What changes for them
The jury	The single most memorable image in the demo, and the one that makes the scale of the output obvious without any explanation.
Non-specialist stakeholders	A way to understand a subsurface product without reading a depth-profile chart.
Ocean scientists	Immediate visual sanity-checking of basin-scale structure that a single profile would hide.
Educators and students	A rotatable, real-data object for teaching how the ocean is layered - built from genuine satellite-driven reconstruction, not an illustration.

WHY SOCIETY GAINS FROM THIS

Public understanding of the ocean is limited by the fact that nobody can see into it. A truthful, rotatable picture built from real reconstruction - one that draws the sea floor as the sea floor - is a genuine public-communication asset for a national ocean agency.

7. Say this to the jury

This is the whole output as one object: every ocean cell, every one of fifteen depths, one day, rotatable. The holes are not missing data - they are the sea floor, and about a quarter of our ocean cells never reach 1000 m. Our classification adds up to exactly 24,000 cells at every depth, which is how we know nothing is being quietly filled in. And the vertical scale is exaggerated, with the factor printed, because at true scale a kilometre of depth across this basin is thinner than paper.

8. Questions the jury will ask

Q. Are those gaps places your model failed?

A. No - they are bathymetry. At 1000 m, 2,859 ocean cells are below the sea floor. We refuse to produce a value there rather than fill it, because a filled cell would assert that water exists where there is rock.

Q. The ocean looks far deeper than it is in that picture.

A. Correct, and that is stated on the page. The vertical axis is exaggerated and the factor is printed. Drawn to true scale the entire 1000 m column would be invisible across sixty degrees of longitude.

Q. Is the science computed at this coarse resolution?

A. No. The reconstruction is on the full 0.25 degree grid - 100 by 240. Only the on-screen sampling is reduced so a browser can render it, and the caption says so.

9. The honest limit - say it before they find it

A PICTURE, NOT A MEASUREMENT

The cube is a rendering of output that is measured elsewhere. No accuracy claim in this project rests on it. Its own honesty checks are structural - that the cell classification sums exactly to the grid and that sea-floor cells are refused rather than filled - and the horizontal sampling shown on screen is deliberately coarser than the science underneath it.

IN ONE LINE

The whole reconstruction as one rotatable body of water, where every hole is the sea floor and the classification adds up to the grid exactly.

24,000

CELLS CLASSIFIED, EXACTLY, AT EVERY DEPTH

11,832

OCEAN CELLS, CONSISTENT ACROSS TWO DEPTHS

~24%

OF OCEAN CELLS NEVER REACH 1000 M

Feature 4 of 17. Next: Feature 5.